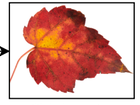


Sample Image

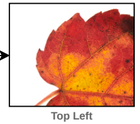
(Arbitrary Resolution)

AnyRes Tiling

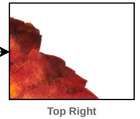
$1 \rightarrow 4(local) + 1(global)$



Global Low Res



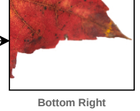
Top Left



Top Right



Bottom Left



Bottom Right

Process Tensor

$[5, 3, 384, 384]$

SigLIP ViT

SO-400M/P14

$[5, 729, 1152]$

Tile Embeddings

(Spatial Grounding)



$tile[i] \rightarrow tile[i] + tile_{embed}[i]$

Vision Projector

$\mathbb{R}^{3645 \times 1152} \rightarrow \mathbb{R}^{3645 \times 3072}$

Latent Queries

$(128, 3072)$

Cross + Self Attention

$Heads = 24$

$Dim = 128$

Feedforward Network

Perceiver Block 1

Cross + Self Attention

$Heads = 24$

$Dim = 128$

Feedforward Network

Perceiver Block 2

Splice & Forward

Perceiver-Resampler

$\mathbb{R}^{3645 \times 3072} \rightarrow \mathbb{R}^{128 \times 3072}$

Decoder Language Model

Microsoft-Phi3-mini (128k-instruct)


```
{
 "diagnosis": "...",
 "type": "fungal",
 "severity": 0.40,
 "confidence": 0.85,
 "symptoms": [...],
 "reasoning": "...",
 "recommended_actions":
     [...]
}
```


Test-Results

(JSON consistent)